

2025 AMC 10B Problem 24 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 24. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 24

Problem:

A frog hops along the number line according to the following rules:

- It starts at 0.
- If it is at 0, then it moves to 1 with probability $\frac{1}{2}$ and disappears with probability $\frac{1}{2}$.
- For $n = 1, 2, 3$, if it is at n , then it moves to $n + 1$ with probability $\frac{1}{4}$, to $n - 1$ with probability $\frac{1}{4}$, and disappears with probability $\frac{1}{2}$.

What is the probability that the frog reaches 4?

Answer Choices:

A. $\frac{1}{101}$

B. $\frac{1}{100}$

C. $\frac{1}{99}$

D. $\frac{1}{98}$

E. $\frac{1}{97}$

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Solution:

Let p_n be the probability that the frog eventually reaches 4 given that it starts at position n .

Once the frog reaches 4, we are done, so

$$p_4 = 1.$$

From 0, the frog either moves to 1 with probability $\frac{1}{2}$ or disappears with probability $\frac{1}{2}$:

$$p_0 = \frac{1}{2}p_1 + \frac{1}{2} \cdot 0 = \frac{1}{2}p_1.$$

For $n = 1, 2, 3$, from position n the frog moves to $n + 1$ with probability $\frac{1}{4}$, to $n - 1$ with probability $\frac{1}{4}$, and disappears with probability $\frac{1}{2}$, so

$$p_n = \frac{1}{4}p_{n+1} + \frac{1}{4}p_{n-1} + \frac{1}{2} \cdot 0 = \frac{1}{4}(p_{n+1} + p_{n-1}).$$

Writing these explicitly for $n = 1, 2, 3$ and using $p_4 = 1$ gives

$$\begin{cases} p_0 = \frac{1}{2}p_1, \\ p_1 = \frac{1}{4}(p_2 + p_0), \\ p_2 = \frac{1}{4}(p_3 + p_1), \\ p_3 = \frac{1}{4}(p_4 + p_2) = \frac{1}{4}(1 + p_2). \end{cases}$$

From $p_0 = \frac{1}{2}p_1$ we have

$$p_0 = \frac{1}{2}p_1.$$

Substitute into the equation for p_1 :

$$p_1 = \frac{1}{4}\left(p_2 + \frac{1}{2}p_1\right) \implies 4p_1 = p_2 + \frac{1}{2}p_1 \implies 8p_1 = 2p_2 + p_1.$$

Thus

$$7p_1 = 2p_2 \implies p_2 = \frac{7}{2}p_1.$$

Use p_2 in the equation for p_2 :

$$p_2 = \frac{1}{4}(p_3 + p_1) \implies 4p_2 = p_3 + p_1.$$

Substitute $p_2 = \frac{7}{2}p_1$:

$$4 \cdot \frac{7}{2}p_1 = p_3 + p_1 \implies 14p_1 = p_3 + p_1 \implies p_3 = 13p_1.$$

Now use the equation for p_3 :

$$p_3 = \frac{1}{4}(1 + p_2) \implies 4p_3 = 1 + p_2.$$

Substitute $p_3 = 13p_1$ and $p_2 = \frac{7}{2}p_1$:

$$4 \cdot 13p_1 = 1 + \frac{7}{2}p_1 \implies 52p_1 = 1 + \frac{7}{2}p_1.$$

We can further simplify with

$$\frac{104}{2}p_1 = 1 + \frac{7}{2}p_1 \implies \left(\frac{104}{2} - \frac{7}{2}\right)p_1 = 1 \implies \frac{97}{2}p_1 = 1.$$

Hence

$$p_1 = \frac{2}{97}.$$

Finally, from $p_0 = \frac{1}{2}p_1$,

$$p_0 = \frac{1}{2} \cdot \frac{2}{97} = \frac{1}{97}.$$

Therefore, the probability that the frog reaches 4 is

(E) $\frac{1}{97}$

Alternate Solution (using a transition table / Markov chain).

We model the frog as a Markov chain with the following states:

$$0, 1, 2, 3, 4, D,$$

where D is a "dead" state (the frog disappeared). We treat 4 and D as absorbing states: once the frog reaches 4 or disappears, it stays there forever for the purposes of this problem.

From the rules of the problem, the one-step transition probabilities are:

From\To	0	1	2	3	4	D
0	0	$\frac{1}{2}$	0	0	0	$\frac{1}{2}$
1	$\frac{1}{4}$	0	$\frac{1}{4}$	0	0	$\frac{1}{2}$
2	0	$\frac{1}{4}$	0	$\frac{1}{4}$	0	$\frac{1}{2}$
3	0	0	$\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{2}$
4	0	0	0	0	1	0
D	0	0	0	0	0	1

Let p_n be the probability that the frog is eventually absorbed in state 4 (i.e., eventually reaches 4) given that it starts in state n .

For the absorbing states:

$$p_4 = 1, \quad p_D = 0.$$

For a non-absorbing state n , the Markov property gives

$$p_n = \sum_{\text{states } s} P(\text{go from } n \text{ to } s \text{ in one step}) \cdot p_s.$$

Using the rows of the transition table:

- From 0: it goes to 1 with probability $\frac{1}{2}$ and to D with probability $\frac{1}{2}$:

$$p_0 = \frac{1}{2}p_1 + \frac{1}{2}p_D = \frac{1}{2}p_1.$$

- From 1: it goes to 0 with probability $\frac{1}{4}$, to 2 with probability $\frac{1}{4}$, and to D with probability $\frac{1}{2}$:

$$p_1 = \frac{1}{4}p_0 + \frac{1}{4}p_2 + \frac{1}{2}p_D = \frac{1}{4}(p_0 + p_2).$$

- From 2:

$$p_2 = \frac{1}{4}p_1 + \frac{1}{4}p_3.$$

- From 3: it goes to 2 with probability $\frac{1}{4}$, to 4 with probability $\frac{1}{4}$, and to D with probability $\frac{1}{2}$:

$$p_3 = \frac{1}{4}p_2 + \frac{1}{4}p_4 + \frac{1}{2}p_D = \frac{1}{4}p_2 + \frac{1}{4}.$$

Thus, we again obtain the system

$$\begin{cases} p_0 = \frac{1}{2}p_1, \\ p_1 = \frac{1}{4}(p_0 + p_2), \\ p_2 = \frac{1}{4}(p_1 + p_3), \\ p_3 = \frac{1}{4}(p_2 + 1). \end{cases}$$

We can regard this as a linear system coming from the transition table, and solve it systematically (for example, by substitution as in Solution 1, or by Gaussian elimination on the augmented matrix). Solving yields

$$p_0 = \frac{1}{97}.$$

Therefore, by the table/Markov-chain method as well, the probability that the frog reaches 4 is

$$\boxed{(E) \frac{1}{97}}$$

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